

Universal Physical Admissibility and Stabilisation Layer (UPASL) Functional-Logical Commit-Gating Across Thermal, Mechanical, Electrical, Radiation, Fluid, and Information Domains

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February 25, 2026

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1 Scope and Objective

This document specifies a **Universal Physical Admissibility and Stabilisation Layer (UPASL)** that makes the *application* of physical laws operationally stable by binding irreversible actions to (i) evidence sufficiency, (ii) domain-specific invariants, (iii) bounded rates, and (iv) cumulative damage budgets.

UPASL does not modify physics. It **stabilizes execution** under limited evidence at irreversibility boundaries (commit events).

Irreversible commit events include, but are not limited to: TX POWER RAMP, high-power mode entry, thruster ignition, slew manoeuvre, deployable release, mode switching, venting, FPGA reconfiguration, and fault recovery actions.

2 Notation, Evidence, and Deterministic Proxies

2.1 Runtime atoms and validity

Let the runtime atom set be $A = \{a_i\}$ with raw measurement $x_i(t)$. Each atom carries age Δt_i and conformance flag $c_i \in \{0, 1\}$.

$$\text{valid}(a_i) := (\Delta t_i \leq \Delta t_i^{\max}) \wedge (c_i = 1). \quad (1)$$

If any required evidence atom is invalid, the corresponding domain state is **UNDETERMINED**.

2.2 Hardware-compatible proxy normalisation

For valid atoms:

$$p_i(t) = \text{clip}\left(\left(\frac{x_i(t) - \mu_i}{\sigma_i}\right) s_i + b_i, 0, 1\right), \quad (2)$$

else $p_i(t) = \perp$ (UNDETERMINED).

Deterministic helper functions

$$\text{clip}(y, 0, 1) = \min\{1, \max\{0, y\}\}, \quad (3)$$

$$\text{sp}(x; \tau) = \tau \ln(1 + e^{x/\tau}), \quad \tau > 0. \quad (4)$$

3 Universal Commit Semantics (Fail-Closed)

Let $\mathcal{C}$ be an irreversible commit action. UPASL produces a deterministic decision:

$$\{\text{ALLOW}, \text{LIMIT}, \text{REFUSE}\}.$$

3.1 Domain statuses

For each domain $d \in \mathcal{D}$ we define:

$$S_d \in \{\text{SAT}, \text{VIOL}, \text{UND}\}.$$

$$S_d = \text{SAT} \Leftrightarrow \bigwedge_k I_{d,k}(t), \quad (5)$$

$$S_d = \text{VIOL} \Leftrightarrow \neg \bigwedge_k I_{d,k}(t), \quad (6)$$

$$S_d = \text{UND} \Leftrightarrow \exists a_i \in A_d : p_i(t) = \perp. \quad (7)$$

3.2 Universal decision rules

Let $h(t)$ be the global hazard score and h_{allow} the admissible bound. Let $\ell(t) \in [0, 1]$ be the global limit fraction.

$$\text{REFUSE} \Leftrightarrow (\exists d : S_d = \text{VIOL}) \vee (\exists d : S_d = \text{UND}), \quad (8)$$

$$\text{ALLOW} \Leftrightarrow (\forall d : S_d = \text{SAT}) \wedge (h(t) \leq h_{\text{allow}}) \wedge (\ell(t) = 1), \quad (9)$$

$$\text{LIMIT} \Leftrightarrow (\forall d : S_d = \text{SAT}) \wedge (h(t) \leq h_{\text{allow}}) \wedge (0 < \ell(t) < 1). \quad (10)$$

Default failure mode is fail-closed:

$$\text{REFUSE}, \quad \ell(t) = 0.$$

4 Universal Structure of Stabilisation

UPASL stabilizes each domain using the same functional schema:

1. **Envelope invariants:** hard safety bounds.
2. **Rate invariants:** bounded derivatives.
3. **Cumulative budget invariants:** bounded integrals (damage/consumption).
4. **Evidence hygiene:** freshness and conformance gates.
5. **Limit operator:** continuous admissible action fraction $\ell_d(t)$.

Global limit fraction:

$$\ell(t) = \min\{1, \ell_T(t), \ell_M(t), \ell_E(t), \ell_R(t), \ell_F(t), \ell_I(t)\}.$$

5 Domain I: Thermodynamics (Thermal)

5.1 State variables

$$T(t) := \text{hotspot temperature}, \quad (11)$$

$$\dot{T}(t) := \frac{dT}{dt}, \quad (12)$$

$$H_T(t) := T_{\max} - T(t) \quad (\text{headroom}). \quad (13)$$

5.2 Invariants

$$I_{T1} : T(t) \leq T_{\max}, \quad (14)$$

$$I_{T2} : |\dot{T}(t)| \leq \dot{T}_{\max}, \quad (15)$$

$$I_{T3} : H_T(t) \geq H_{T,\min}. \quad (16)$$

5.3 Cycling / cumulative stress (bounded)

$$A_T(t) = \max_{s \in [t-W, t]} T(s) - \min_{s \in [t-W, t]} T(s), \quad (17)$$

$$\dot{C}_T(t) = \lambda_T \cdot \text{sp}(A_T(t) - A_{T,\max}; \tau_T) - \omega_T C_T(t). \quad (18)$$

$$I_{T4} : C_T(t) \leq C_{T,\max}. \quad (19)$$

5.4 Limit fraction

$$\ell_T(t) = \min\{\ell_{T,H}(t), \ell_{T,\dot{T}}(t), \ell_{T,C}(t)\}, \quad (20)$$

$$\ell_{T,H}(t) = \text{clip}\left(\frac{H_T(t) - H_{T,\min}}{H_T - H_{T,\min}}, 0, 1\right), \quad (21)$$

$$\ell_{T,\dot{T}}(t) = \text{clip}\left(1 - \frac{|\dot{T}(t)|}{\dot{T}_{\max}}, 0, 1\right), \quad (22)$$

$$\ell_{T,C}(t) = \text{clip}\left(1 - \frac{C_T(t)}{C_T}, 0, 1\right). \quad (23)$$

6 Domain II: Mechanics (Dynamics / Structures)

6.1 State variables

Let:

$$\sigma(t) := \text{representative structural stress / load proxy}, \quad (24)$$

$$\dot{\sigma}(t) := \frac{d\sigma}{dt}, \quad (25)$$

$$m_L(t) := \text{load margin (positive means safe)}. \quad (26)$$

6.2 Invariants

$$I_{M1} : m_L(t) \geq m_{L,\min}, \quad (27)$$

$$I_{M2} : |\dot{\sigma}(t)| \leq \dot{\sigma}_{\max}. \quad (28)$$

6.3 Fatigue / cumulative damage (bounded)

Let cumulative mechanical damage be:

$$D_M(t) = \int_0^t \max\{0, \sigma(\tau) - \sigma_{\text{safe}}\} d\tau, \quad (29)$$

with invariant:

$$I_{M3} : D_M(t) \leq D_{M,\cdot}. \quad (30)$$

6.4 Limit fraction

$$\ell_M(t) = \min\{\ell_{M,m}(t), \ell_{M,\dot{\sigma}}(t), \ell_{M,D}(t)\}, \quad (31)$$

$$\ell_{M,m}(t) = \text{clip}\left(\frac{m_L(t) - m_{L,\min}}{m_L - m_{L,\min}}, 0, 1\right), \quad (32)$$

$$\ell_{M,\dot{\sigma}}(t) = \text{clip}\left(1 - \frac{|\dot{\sigma}(t)|}{\dot{\sigma}_{\max}}, 0, 1\right), \quad (33)$$

$$\ell_{M,D}(t) = \text{clip}\left(1 - \frac{D_M(t)}{D_{M,\cdot}}, 0, 1\right). \quad (34)$$

7 Domain III: Electrodynamics (Power / EPS)

7.1 State variables

$$V(t) := \text{bus voltage}, \quad (35)$$

$$I(t) := \text{bus current}, \quad (36)$$

$$\dot{I}(t) := \frac{dI}{dt}, \quad (37)$$

$$\text{SoC}(t) := \text{battery state-of-charge}, \quad (38)$$

$$m_P(t) := \frac{P_{\text{avail}}(t)}{P_{\text{need}}(t)} \quad (\text{power margin}). \quad (39)$$

7.2 Invariants

$$I_{E1} : V(t) \geq V_{\min}, \quad (40)$$

$$I_{E2} : |\dot{I}(t)| \leq \dot{I}_{\max}, \quad (41)$$

$$I_{E3} : \text{SoC}(t) \geq \text{SoC}_{\min}, \quad (42)$$

$$I_{E4} : m_P(t) \geq m_{P,\min}. \quad (43)$$

7.3 Energy budget (bounded)

$$E_{\text{draw}}(t) = \int_0^t P(\tau) d\tau, \quad I_{E5} : E_{\text{draw}}(t) \leq E. \quad (44)$$

7.4 Limit fraction

$$\ell_E(t) = \min\{\ell_{E,V}(t), \ell_{E,I}(t), \ell_{E,\text{SoC}}(t), \ell_{E,m}(t)\}, \quad (45)$$

$$\ell_{E,V}(t) = \text{clip}\left(\frac{V(t) - V_{\min}}{V - V_{\min}}, 0, 1\right), \quad (46)$$

$$\ell_{E,I}(t) = \text{clip}\left(1 - \frac{|\dot{I}(t)|}{\dot{I}_{\max}}, 0, 1\right), \quad (47)$$

$$\ell_{E,\text{SoC}}(t) = \text{clip}\left(\frac{\text{SoC}(t) - \text{SoC}_{\min}}{\text{SoC} - \text{SoC}_{\min}}, 0, 1\right), \quad (48)$$

$$\ell_{E,m}(t) = \text{clip}\left(\frac{m_P(t) - m_{P,\min}}{m_P - m_{P,\min}}, 0, 1\right). \quad (49)$$

8 Domain IV: Radiation (Dose / SEE Risk)

8.1 State variables

$$\dot{D}(t) := \text{dose rate proxy}, \quad (50)$$

$$D(t) := \int_0^t \dot{D}(\tau) d\tau \quad (\text{cumulative dose}), \quad (51)$$

$$r_{\text{SEE}}(t) := \text{SEE rate proxy}. \quad (52)$$

8.2 Invariants

$$I_{R1} : D(t) \leq D, \quad (53)$$

$$I_{R2} : \dot{D}(t) \leq \dot{D}_{\max}, \quad (54)$$

$$I_{R3} : r_{\text{SEE}}(t) \leq r_{\text{SEE},\max}. \quad (55)$$

8.3 Limit fraction

$$\ell_R(t) = \min\{\ell_{R,D}(t), \ell_{R,\dot{D}}(t), \ell_{R,\text{SEE}}(t)\}, \quad (56)$$

$$\ell_{R,D}(t) = \text{clip}\left(1 - \frac{D(t)}{D}, 0, 1\right), \quad (57)$$

$$\ell_{R,\dot{D}}(t) = \text{clip}\left(1 - \frac{\dot{D}(t)}{\dot{D}_{\max}}, 0, 1\right), \quad (58)$$

$$\ell_{R,\text{SEE}}(t) = \text{clip}\left(1 - \frac{r_{\text{SEE}}(t)}{r_{\text{SEE},\max}}, 0, 1\right). \quad (59)$$

9 Domain V: Fluid / Two-Phase / Slosh Regimes

9.1 State variables

Let:

$$\text{fill}(t) := \text{tank fill fraction}, \quad (60)$$

$$\tau_s(t) := \text{slosh torque proxy}, \quad (61)$$

$$\Pi(t) := \text{regime proximity index (phase-change / instability)}. \quad (62)$$

9.2 Invariants

$$I_{F1} : \tau_s(t) \leq \tau_{s,\max}, \quad (63)$$

$$I_{F2} : \Pi(t) \leq \Pi_{\max}, \quad (64)$$

$$I_{F3} : \text{fill}(t) \notin \mathcal{Z}_{\text{unstable}} \quad (\text{avoid unstable fill zones}). \quad (65)$$

9.3 Limit fraction

$$\ell_F(t) = \min\{\ell_{F,\tau}(t), \ell_{F,\Pi}(t)\}, \quad (66)$$

$$\ell_{F,\tau}(t) = \text{clip}\left(1 - \frac{\tau_s(t)}{\tau_{s,\max}}, 0, 1\right), \quad (67)$$

$$\ell_{F,\Pi}(t) = \text{clip}\left(1 - \frac{\Pi(t)}{\Pi_{\max}}, 0, 1\right). \quad (68)$$

10 Domain VI: Information / Timing (Evidence Physics)

10.1 State variables

Let:

$$\Delta t_{\max}^{(d)} := \text{maximum admissible evidence age for domain } d, \quad (69)$$

$$L(t) := \text{control-loop latency}, \quad (70)$$

$$J(t) := \text{jitter / timing instability proxy}. \quad (71)$$

10.2 Invariants

$$I_{I1} : \quad \forall a_i \in A_{\text{crit}} : \Delta t_i \leq \Delta t_{\text{max}}^{(d(i))}, \quad (72)$$

$$I_{I2} : \quad L(t) \leq L_{\text{max}}, \quad (73)$$

$$I_{I3} : \quad J(t) \leq J_{\text{max}}. \quad (74)$$

10.3 Limit fraction

$$\ell_I(t) = \min\{\ell_{I,L}(t), \ell_{I,J}(t)\}, \quad (75)$$

$$\ell_{I,L}(t) = \text{clip}\left(1 - \frac{L(t)}{L_{\text{max}}}, 0, 1\right), \quad (76)$$

$$\ell_{I,J}(t) = \text{clip}\left(1 - \frac{J(t)}{J_{\text{max}}}, 0, 1\right). \quad (77)$$

11 Global Hazard and Functional-Logical Stability

UPASL integrates domain violations into a global hazard score. A generic saturated hazard form:

$$h(t) = h_{\text{max}} (1 - \exp(-(k_1 D_{\text{band}} + k_2 O + k_3 \Gamma + k_4 D_{\text{in}})))$$

with Γ including rate contributions from all domains.

11.1 Stability operator (Trias form)

$$\Phi^*(t) = w_G G^*(t) + w_R R^*(t) + w_M M^*(t), \quad w_G + w_R + w_M = 1.$$

Calibration follows a **dynamic corridor** (approx. 60–80% freedom to 40–20% order) as a calibration range, not a fixed ratio.

12 Implementation Notes (Determinism and Non-Bypassability)

- All computations must be deterministic and suitable for fixed-point implementation.
- All domain decisions must be non-bypassable at the actuator commit boundary.
- Evidence invalidity must trigger fail-closed behavior.
- LIMIT output must be physically enforceable (e.g. power enable line, throttle DAC, command gate).

13 License

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